

1 Important Constants & Equations

This week you covered the concept of electric potential and electric potential energy. This will lead very nicely into the analysis of circuits in chapters 27 to 32!

The **electric potential energy** between two parallel-plate capacitors, very similar to gravitational potential energy ($U_g = mgh$) is defined as:

$$U_{\text{elec}} = qEs \quad (\text{Electric Potential Energy in a Uniform Electric Field})$$

There is also an electric potential energy for a system of two point charges:

$$U_{\text{elec}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \quad (\text{Potential Energy for a System of Two Particles})$$

The **electric potential** is a property of the source charge and exists without the presence of another charge to 'feel' it:

$$V \equiv \frac{U_{\text{q + sources}}}{q} \implies U_{\text{elec}} = qV \quad (\text{Relationship Between Potential and Potential Energy})$$

Given the above equation, it shouldn't be surprising that the parallel-plate capacitor of length d (which generates an electric field) has an associated voltage:

$$E = \frac{\Delta V_C}{d} \quad (\text{Electric Field of a Parallel-Plate Capacitor})$$

Lastly, a point charge also has an associated voltage at a distance r . It is important to note that the potential is a *scalar*, NOT a vector:

$$V_{\text{point}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (\text{Electric Potential of a Point Charge})$$

The following constants/relations are important for the study of electricity:

$$\begin{array}{lll} e = 1.60 \times 10^{-19} \text{ C} & (\text{Fundamental Unit of Charge}) & m_p = 1.67 \times 10^{-27} \text{ kg} \quad (\text{Mass of Proton}) \\ m_e = 9.11 \times 10^{-31} \text{ kg} & (\text{Mass of Electron}) & K = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2 \quad (\text{Electrostatic Constant}) \\ \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 & (\text{Permittivity Constant}) & \lambda = dq/dl \quad \eta = dq/dA \quad \rho = dq/dV \end{array}$$

2 Problems

1. Though confusing, *electric potential energy* and the *electric potential* are two different concepts, but they are closely related. What is the difference between the *electric potential energy* and the *electric potential*?

2. A particle with a mass $m = 0.1\text{g}$ and a charge $q = 5\text{nC}$ and moves with a speed of $v_i = 2 \times 10^6 \text{ m/s}$ through an electric potential of 0V . The particle then encounters a potential of -20V .

(a) What is the potential difference ΔV of this system?

$$\Delta V =$$

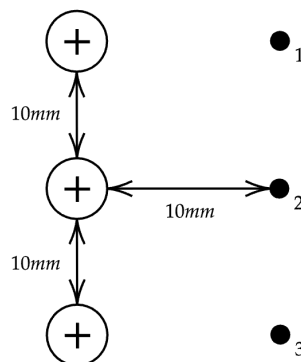
(b) Recall that the kinetic energy of a particle is $K = \frac{1}{2}mv^2$ and the potential energy of a charged particle in a potential V is $U = qV$. Write an expression for the initial and final energies of system.

(c) Find an expression for v_f in terms of m , q , ΔV , and v_i .

(d) Find v_f .

(e) **(BONUS)**. What if ΔV was $+20\text{V}$? What would v_f be then?

3. Below is a system where three equally-spaced particles with charge $q = 5\text{nC}$ are placed along the y-axis. In addition, three test points have been added.



(a) Determine the electric potential at position 1.

(b) Determine the electric potential at position 2.

(c) Determine the electric potential at position 3.

4. (**CHALLENGE PROBLEM**). A charged sphere with radius R has a charge density of $\rho = 3r^2$, where $0 < r \leq R$ is the distance from the center of the sphere.

(a) Diagram this scenario, clearly indicating that the density ρ is increasing as $r \rightarrow R$.

(b) Find the total charge Q of the sphere. (Hint: Recall that $\rho = \frac{dq}{dV}$)

(c) Find the electric potential V a distance z from the *edge* of the sphere.

(d) Find the electric field E a distance z from the *edge* of the sphere.

(e) Using the values for E and V you found in parts (b) and (c), respectively, determine the distance of separation d a parallel-plate capacitor would have to be to generate such an electric field E and potential V .

Assume that one of the plates has a potential of $V_1 = 0$ and the other plate has a potential of $V_2 = V$, meaning $\Delta V_C = V_2 - V_1 = V$.

(f) **(CHALLENGE BONUS)**. Now let the sphere be hollow, with an inner radius of R_{in} and an outer radius of R_{out} . What would be the electric potential a distance $x < R_{\text{in}}$ from the center of the sphere. (Hint: The charge density is the same, $\rho = 3r^2$, but the charge will be different)